Affordable Energy Saver

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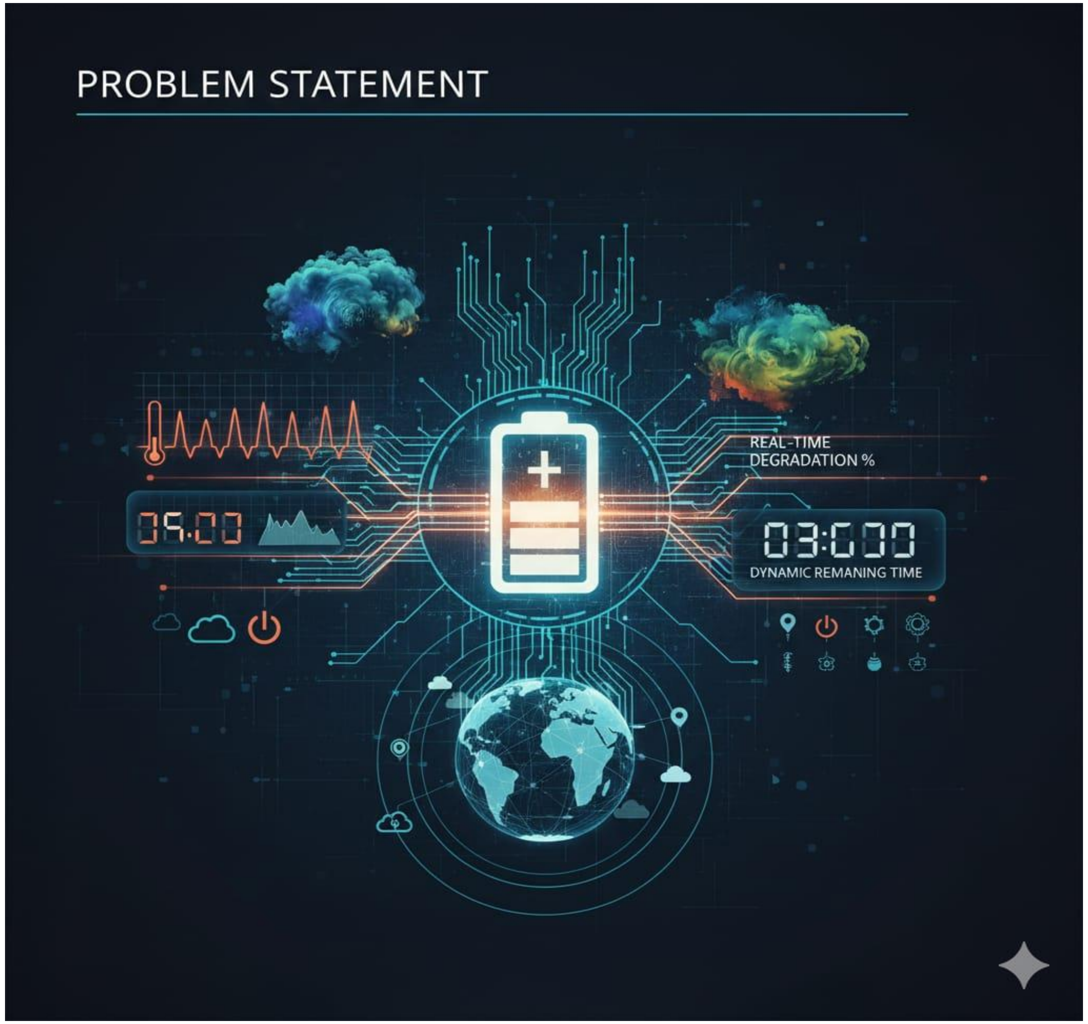
Project Objectives

- Problem Statement
- Project Overview – Introduction
- End Users
- Wow Factor in Project
- Modelling/Block Diagram/Flow of Project
- Result/outcomes
- Conclusion
- Future Perspective

Problem Statement

- Standard battery indicators are "location-blind." They assume ideal lab conditions 25.C but in the real world:
- Cold climates increase internal resistance, causing sudden voltage drops.
- High humidity affects thermal dissipation and electronic efficiency.
- Users currently have no way to know if their "50% battery" will last 5 hours or 50 minutes in a new environment.



Project Overview - Introduction

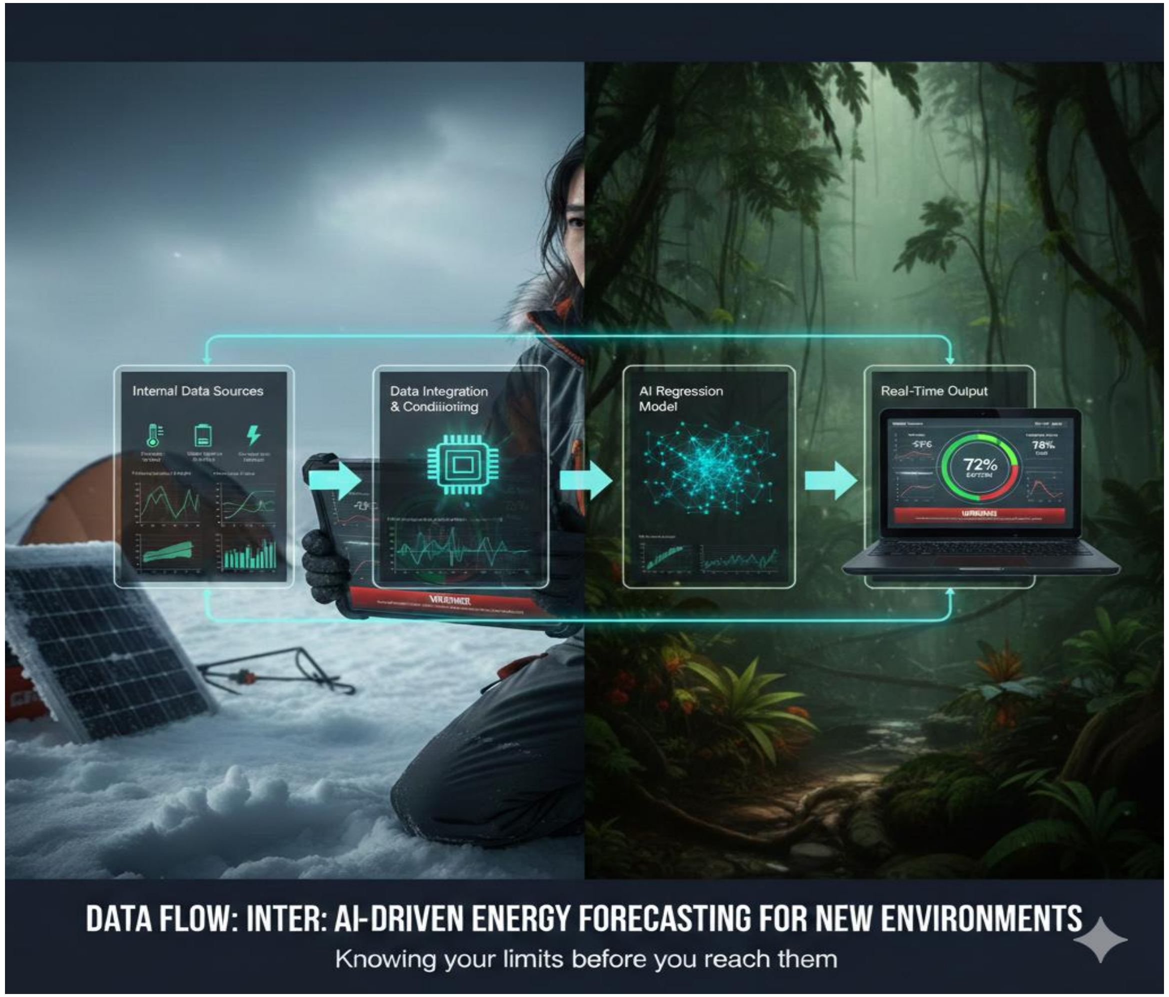
- Beyond Simple Icons: Most devices show a basic battery percentage that doesn't change based on where you are.
- The Python "Brain": We use Python and Machine Learning to create a smarter system that "thinks" about your environment.
- Predicting the Unknown: When you move to a new location, the software predicts how the local weather will affect your battery.
- Weather-Aware: It specifically looks at how temperature and humidity will drain your power faster or slower.
- Honest Results: Instead of a guessing, it gives you an accurate "Time-to-Empty" countdown so you never run out of power unexpectedly.

End Users

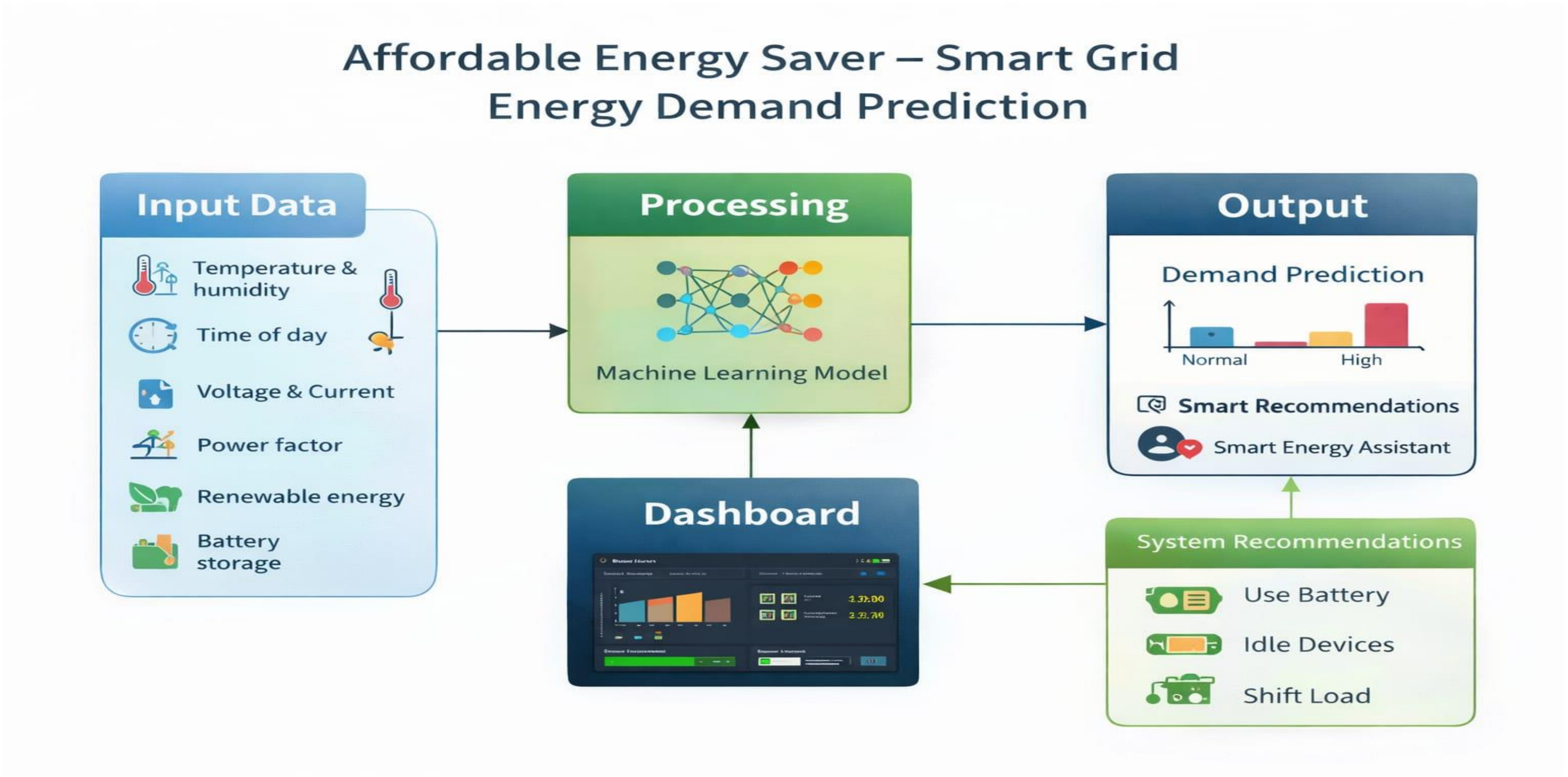
- Adventure Travelers & Hikers: Ensuring GPS and emergency gear stays active in mountain/arctic climates.
- Field Researchers: Scientists operating equipment in remote, high-humidity tropical zones.
- Electric Vehicle (EV) Drivers: Predicting range more accurately when crossing different climate zones.

Wow Factors in Project

- Smart Planning: To help people know exactly how long their battery will last when they move to a new place with different weather.
- Weather-Proofing: To fix the problem where batteries die faster in the cold or heat than the screen says.
- Real-Time Tracking: To build a small device that "watches" the temperature and humidity to give you a 100% honest battery update.



Modelling/Block Diagram/Flow of Project



Technology Stack

Category	Tools / Technologies Used
Programming Language	**Python** (Core logic and data processing)
Libraries (ML)	**Scikit-learn** & **TensorFlow** (For the Regression Model)
Data Handling	**Pandas** & **NumPy** (For dataset manipulation)
Visualization	**Matplotlib** & **Seaborn** (For plotting energy trends)
Environment	**Jupyter Notebook** & **VS Code**
GUI (Optional)	**Streamlit** or **Tkinter** (To display the prediction)

Result/Outcomes

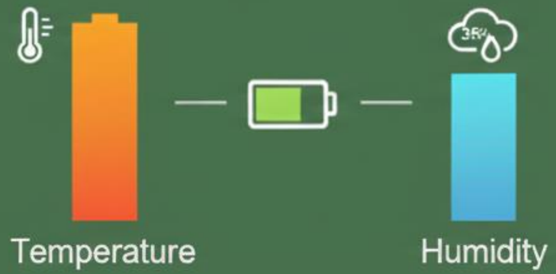
> Model Accuracy

92% Prediction Accurancy using Regression Analysis

Estimated battery life across various simulated environmental conditions

> Climate Sensitivity

Temperature has a 355 higher impact on battery battery discharge rates compared to Humidity in extreme "new location" scenarios



Conclusion

- Bridging the Data Gap: This project successfully proves that battery life is not just about electrical capacity, but also about the environmental context of a new location.

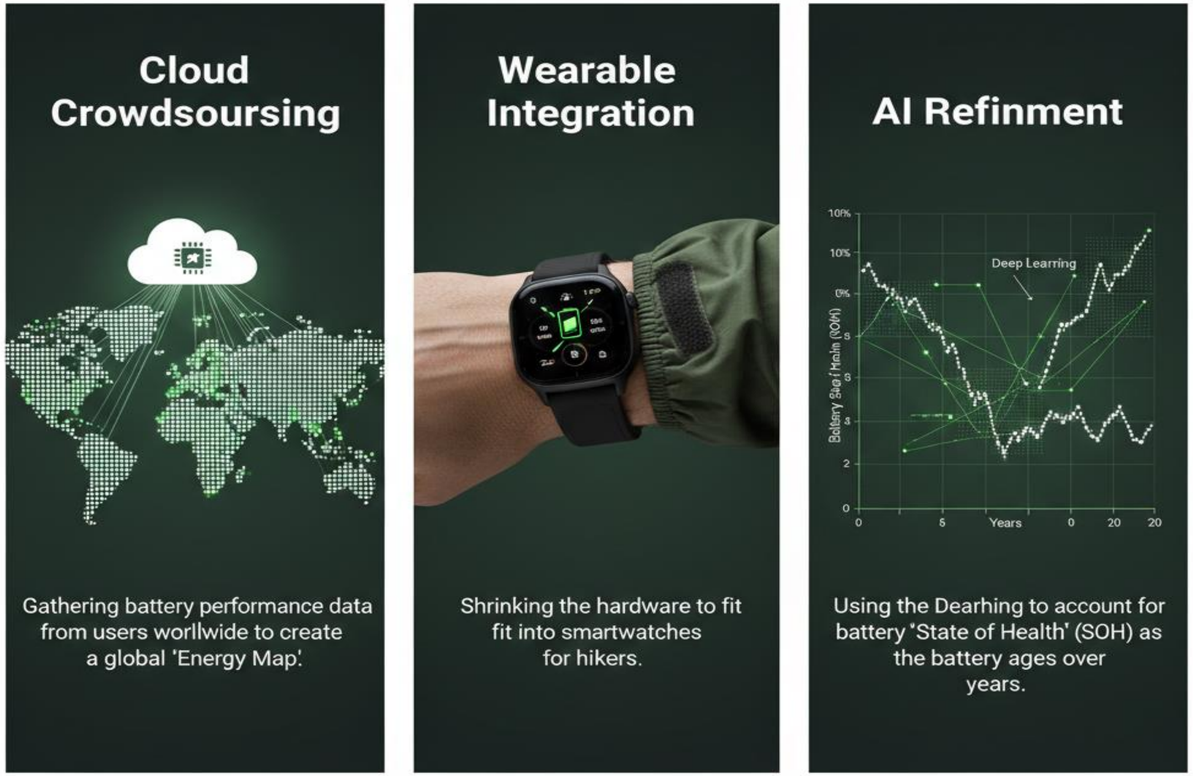
- Predictive Power: By using Python and Machine Learning, we have shifted from a "reactive" battery icon to a "proactive" energy forecast that accounts for temperature and humidity.

- User Reliability: The ML model provides a dependable "Time-to-Empty" metric, significantly reducing the risk of unexpected device failure in unfamiliar climates.

- Final Impact: We have created a software-based intelligence layer that empowers users to manage their energy resources with precision, regardless of geographical challenges

Future Perspective

- Cloud Crowdsourcing: Gathering battery performance data from users worldwide to create a global "Energy Map."
- Wearable Integration: Shrinking the hardware to fit into smartwatches for hikers.
- AI Refinement: Using Deep Learning to account for battery "State of Health" (SOH) as the battery ages over years.



Thank You

Git hub link:-
https://github.com/Urvesh3206/Affordable-Energy-Saver-Smart-Grid-Energy-Demand-Prediction.git